

Multi-Model Neural CDE Interpretability Audit Report

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Models Compared: 3

Samples Used: 3000

Seed: 42

Appendix A: Mathematical Foundations

This appendix provides the mathematical basis for the interpretability metrics used throughout this audit.

A.1 Neural CDE Architecture

The CondorBrain uses **Neural Controlled Differential Equations** for continuous-time market dynamics modeling.

Core CDE Equation:

$$dZ_t = f(Z_t; \theta) dX_t$$

Where $Z_t \in R^H$ is the latent state and $f: R^H \rightarrow R^{H \times D}$ is the learned vector field.

Integral Form:

$$Z_T = Z_0 + \int_0^T f(Z_t) dX_t$$

Explicit Euler Discretization:

$$Z_{t+1} = Z_t + f(Z_t) \cdot (X_{t+1} - X_t)$$

Vector Field Network (Stability-Bounded):

$$f(z) = \tanh(W_2 \cdot \text{SiLU}(W_1 \cdot z + b_1) + b_2)$$

The tanh activation ensures $\|f(z)\|_{\infty} \leq 1$ preventing state explosion over long sequences.

Interpretation:

All components of the vector field are bounded in magnitude by 1. This enforces a **Lipschitz-like stability condition**, ensuring that state updates $dZ_t = f(Z_t) dX_t$ cannot explode even when control increments are large, thereby stabilizing long-horizon Neural CDE integration.

A.2 Permutation Importance

Feature importance is computed by measuring prediction degradation when features are randomly shuffled:

$$I_j = \frac{1}{K} \sum_{k=1}^K [\mathcal{L}(f(X^{(j,k)}), y) - \mathcal{L}(f(X), y)]$$

Where $X^{(j,k)}$ denotes data with feature j permuted in repetition k .

Normalization:

$$\hat{I}_j = \frac{I_j}{\sum_{i=1}^D I_i} \times 100\%$$

A.3 Divergence Metrics

Cosine Similarity:

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

Jensen-Shannon Divergence:

$$D_{JS}(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M)$$

Where $M = \frac{1}{2}(P + Q)$ and $D_{KL}(P \parallel Q)$ is the Kullback-Leibler divergence:

$$D_{KL}(P \parallel Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

Wasserstein Distance (Earth Mover's):

$$W_1(P, Q) = \inf_{\gamma \in \Gamma(P, Q)} \mathbb{E}_{(x, y) \sim \gamma} [\|x - y\|]$$

For 1D distributions, this simplifies to the area between CDFs:

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| dx$$

A.4 Gradient Saliency

Saliency maps measure input sensitivity via backpropagated gradients:

$$S_j = \frac{1}{N} \sum_{i=1}^N \left| \frac{\partial \hat{y}^{(i)}}{\partial x_j^{(i)}} \right|$$

Temporal Saliency (across sequence positions):

$$S_t = \sum_{j=1}^D \left| \frac{\partial \hat{y}}{\partial x_{t,j}} \right|$$

Gradient Alignment (between models A and B):

$$\rho_{grad} = \frac{\mathbf{g}_A \cdot \mathbf{g}_B}{\|\mathbf{g}_A\| \|\mathbf{g}_B\|}$$

A.5 Fisher Information Matrix

The Fisher Information Matrix captures parameter sensitivity:

$$\mathcal{F}_{ij} = \mathbb{E} \left[\frac{\partial \log p(y | x; \theta)}{\partial \theta_i} \frac{\partial \log p(y | x; \theta)}{\partial \theta_j} \right]$$

Empirical Estimation (diagonal approximation):

$$\hat{\mathcal{F}}_{jj} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\partial \mathcal{L}^{(i)}}{\partial \theta_j} \right)^2$$

Fisher Overlap (between models):

$$\text{Overlap} = \frac{\mathbf{f}_A \cdot \mathbf{f}_B}{\|\mathbf{f}_A\| \|\mathbf{f}_B\|}$$

A.6 Hessian Eigenspectrum

The Hessian matrix of the loss landscape reveals optimization geometry:

$$H_{ij} = \frac{\partial^2 \mathcal{L}}{\partial \theta_i \partial \theta_j}$$

Eigenvalue Interpretation:

- $\lambda_{\max} > 0$: Maximum curvature (sharpness of minimum)
- $\text{Tr}(H) = \sum_i \lambda_i$: Average curvature
- $\frac{\lambda_{\max}}{\lambda_{\min}}$: Condition number (training stability)

Flatness Correlation:

Flatter minima (lower $\lambda_{\max}$) correlate with better generalization.

A.7 Stability Physics

Mean Squared Deviation (MSD):

$$\text{MSD}(A, B) = \frac{1}{N} \sum_{i=1}^N (y_A^{(i)} - y_B^{(i)})^2$$

Variance Ratio:

$$\rho_{var} = \frac{\text{Var}(y_A)}{\text{Var}(y_B)}$$

Values near 1.0 indicate similar output distributions; deviations suggest regime sensitivity differences.

Kolmogorov-Smirnov Statistic:

$$D_{KS} = \sup_x |F_A(x) - F_B(x)|$$

Maximum difference between empirical CDFs, sensitive to both location and shape differences.

1. Executive Summary

Best Model: epoch_3_013026
Overall Health: 12 strengths identified, 9 concerns flagged

Key finding: feature attribution, gradients, and information geometry are **highly conserved across retraining**, while output heads remain intentionally decorrelated.

2. Configuration & Preprocessing Contract

Feature Registry

- **Input Dimension:** 54
- **Feature Schema:** V2.2 (FEATURE_COLS_V22)
- **Total Features:** 54

Sequence Configuration

- epoch_3_013026: seq_len = 240
- epoch_4_013026: seq_len = 240
- epoch_5_013026: seq_len = 240

Leakage Masking

- target_spot -> 0
- max_dd_60m -> 0

Scaling

- Method: Robust Scaling (median/MAD)
- Clip Range: [-10, +10]
- NaN/Inf Handling: Replace with 0

3. Per-Model Interpretability

epoch_3_013026

All Features (Permutation Importance):

Rank	Feature	Importance (%)
1	ivr	11.88
2	theta	9.55
3	delta	8.47
4	pressure_down	6.25
5	gap_risk_score	5.12
6	pressure_up	4.89
7	bw_expansion_rate	4.49
8	macd_histogram	4.35
9	macd_norm	3.73
10	atr_pct	3.13
11	macd_signal_norm	3.12
12	plus_di	2.35
13	bb_sigma_dyn	2.20
14	vol_ewma	2.00
15	psar_trend	1.97
16	gamma	1.96
17	minus_di	1.91
18	mtf_consensus	1.76
19	vega	1.69
20	te	1.46
21	log_return	1.43

Rank	Feature	Importance (%)
22	kappa_proxy	1.33
23	ret_z	1.30
24	strike	1.27
25	vol_energy	1.08
26	bb_percentile	1.07
27	friction_ratio	1.03
28	breakout_score	1.01
29	consolidation_score	0.93
30	fuzzy_reversion_11	0.89
31	cmf	0.81
32	adx_adaptive	0.69
33	open	0.65
34	stoch_k_dyn	0.57
35	close	0.56
36	psar_adaptive	0.56
37	rsi_dyn	0.49
38	bb_mu_dyn	0.49
39	bb_upper_dyn	0.44
40	bb_lower_dyn	0.36
41	high	0.22
42	low	0.18
43	chaos_membership	0.16
44	position_size_mult	0.16
45	exec_allow	0.02
46	volume	0.00
47	iv	0.00
48	spread_ratio	0.00
49	target_spot	0.00
50	max_dd_60m	0.00
51	risk_override	0.00
52	iv_confidence	0.00
53	psar_reversion_mu	0.00
54	beta1_norm_stub	0.00

Surrogate Tree R2: 0.6814 (High interpretability)

Output Statistics (ROI Head):

- Mean: -0.0591, Std: 0.0723
- Skewness: 0.597, Kurtosis: -1.213

epoch_4_013026

All Features (Permutation Importance):

Rank	Feature	Importance (%)
1	ivr	16.19
2	theta	10.31
3	delta	9.98
4	pressure_down	5.87
5	gap_risk_score	4.57
6	pressure_up	4.07
7	macd_histogram	4.03
8	bw_expansion_rate	3.95
9	macd_norm	3.29
10	atr_pct	2.85
11	macd_signal_norm	2.84
12	plus_di	2.05
13	minus_di	1.97
14	gamma	1.70
15	vega	1.64
16	psar_trend	1.59
17	te	1.46
18	mtf_consensus	1.46
19	strike	1.36
20	bb_sigma_dyn	1.32
21	breakout_score	1.26
22	log_return	1.22
23	ret_z	1.20
24	vol_energy	1.03
25	kappa_proxy	1.02
26	bb_percentile	0.98
27	vol_ewma	0.95
28	fuzzy_reversion_11	0.94
29	open	0.91

Rank	Feature	Importance (%)
30	consolidation_score	0.90
31	adx_adaptive	0.81
32	friction_ratio	0.80
33	cmf	0.77
34	close	0.72
35	stoch_k_dyn	0.64
36	bb_mu_dyn	0.61
37	rsi_dyn	0.55
38	bb_upper_dyn	0.53
39	psar_adaptive	0.52
40	bb_lower_dyn	0.42
41	high	0.31
42	low	0.21
43	chaos_membership	0.11
44	position_size_mult	0.09
45	exec_allow	0.01
46	volume	0.00
47	iv	0.00
48	spread_ratio	0.00
49	target_spot	0.00
50	max_dd_60m	0.00
51	risk_override	0.00
52	iv_confidence	0.00
53	psar_reversion_mu	0.00
54	beta1_norm_stub	0.00

Surrogate Tree R2: 0.7477 (High interpretability)

Output Statistics (ROI Head):

- Mean: -0.0603, Std: 0.0776
- Skewness: 0.740, Kurtosis: -0.871

epoch_5_013026

All Features (Permutation Importance):

Rank	Feature	Importance (%)
1	ivr	17.55
2	theta	11.73
3	delta	11.05
4	pressure_down	5.87
5	pressure_up	4.17
6	gap_risk_score	4.05
7	macd_histogram	3.97
8	bw_expansion_rate	3.89
9	macd_norm	3.14
10	macd_signal_norm	2.82
11	atr_pct	2.60
12	minus_di	2.01
13	plus_di	1.71
14	gamma	1.66
15	psar_trend	1.65
16	vega	1.59
17	te	1.33
18	mtf_consensus	1.28
19	strike	1.20
20	ret_z	1.19
21	log_return	1.14
22	bb_sigma_dyn	1.08
23	vol_energy	1.02
24	breakout_score	0.99
25	kappa_proxy	0.99
26	bb_percentile	0.88
27	fuzzy_reversion_11	0.87
28	friction_ratio	0.84
29	open	0.77
30	consolidation_score	0.77
31	cmf	0.74
32	adx_adaptive	0.72
33	vol_ewma	0.67
34	close	0.66
35	stoch_k_dyn	0.58
36	bb_mu_dyn	0.52
37	rsi_dyn	0.51

Rank	Feature	Importance (%)
38	psar_adaptive	0.51
39	bb_upper_dyn	0.42
40	bb_lower_dyn	0.30
41	high	0.24
42	low	0.18
43	chaos_membership	0.07
44	position_size_mult	0.04
45	exec_allow	0.01
46	volume	0.00
47	iv	0.00
48	spread_ratio	0.00
49	target_spot	0.00
50	max_dd_60m	0.00
51	risk_override	0.00
52	iv_confidence	0.00
53	psar_reversion_mu	0.00
54	beta1_norm_stub	0.00

Surrogate Tree R2: 0.8012 (High interpretability)

Output Statistics (ROI Head):

- Mean: -0.0591, Std: 0.0800
- Skewness: 0.767, Kurtosis: -0.759

4. Pairwise Divergence Analysis

epoch_3_013026 vs epoch_4_013026

Similarity Metrics:

- Cosine Similarity: 0.9829
- Spearman Correlation: 0.9840 (p=1.25e-40)
- Kendall Tau: 0.9254

Distance Metrics:

- Jensen-Shannon Divergence: 0.0716
- Wasserstein Distance: 0.0028

Top-K Feature Overlap:

- Top-5 Overlap: 100%
- Top-10 Overlap: 100%

Interpretation:

- Near-identical importance direction (cosine=0.983): models learned same feature weighting
- Minimal information distance (JS=0.072): importance distributions nearly identical
- Rank order preserved (Spearman=0.984): feature priority ordering unchanged

All Importance Shifts (sorted by magnitude):

Feature	Shift (%)
ivr	-4.32
delta	-1.51
vol_ewma	+1.05
bb_sigma_dyn	+0.88
pressure_up	+0.82
theta	-0.76
gap_risk_score	+0.55
bw_expansion_rate	+0.54
macd_norm	+0.44
pressure_down	+0.39

epoch_3_013026 vs epoch_5_013026

Similarity Metrics:

- Cosine Similarity: 0.9729
- Spearman Correlation: 0.9772 (p=1.16e-36)
- Kendall Tau: 0.9211

Distance Metrics:

- Jensen-Shannon Divergence: 0.0955
- Wasserstein Distance: 0.0039

Top-K Feature Overlap:

- Top-5 Overlap: 80%
- Top-10 Overlap: 90%

Interpretation:

- Near-identical importance direction (cosine=0.973): models learned same feature weighting
- Minimal information distance (JS=0.096): importance distributions nearly identical
- Rank order preserved (Spearman=0.977): feature priority ordering unchanged

All Importance Shifts (sorted by magnitude):

Feature	Shift (%)
ivr	-5.67
delta	-2.58
theta	-2.18
vol_ewma	+1.33
bb_sigma_dyn	+1.12
gap_risk_score	+1.06
pressure_up	+0.72
plus_di	+0.64
bw_expansion_rate	+0.61
macd_norm	+0.58

epoch_4_013026 vs epoch_5_013026

Similarity Metrics:

- Cosine Similarity: 0.9977
- Spearman Correlation: 0.9965 (p=1.17e-57)
- Kendall Tau: 0.9699

Distance Metrics:

- Jensen-Shannon Divergence: 0.0368
- Wasserstein Distance: 0.0014

Top-K Feature Overlap:

- Top-5 Overlap: 80%
- Top-10 Overlap: 90%

Interpretation:

- Near-identical importance direction (cosine=0.998): models learned same feature weighting

- Minimal information distance (JS=0.037): importance distributions nearly identical
- Rank order preserved (Spearman=0.996): feature priority ordering unchanged

All Importance Shifts (sorted by magnitude):

Feature	Shift (%)
theta	-1.42
ivr	-1.36
delta	-1.07
gap_risk_score	+0.51
plus_di	+0.34
vol_ewma	+0.28
breakout_score	+0.27
atr_pct	+0.24
bb_sigma_dyn	+0.24
mtf_consensus	+0.17

5. Stability Analysis (Physics-Inspired)

epoch_3_013026 vs epoch_4_013026

Head	Correlation	MSD	Var Ratio	KS Stat	Wasserstein
call_offset	0.011	0.1749	0.677	0.111	0.0460
put_offset	0.012	0.1671	0.643	0.112	0.0469
wing_width	0.012	0.7912	0.825	0.105	0.0810
dte	0.012	36.0767	0.767	0.128	0.5683
pop	0.010	0.0004	6.894	0.262	0.0091
roi	0.012	0.0111	0.867	0.157	0.0077
max_loss	0.010	0.0322	1.288	0.107	0.0170
confidence	0.000	0.0046	0.796	0.194	0.0143
entry	0.002	0.4145	0.563	0.221	0.2102
exit	0.002	0.4546	1.332	0.096	0.0906

Physics Interpretation:

- Low coherence (avg correlation=0.008): models responding differently to same inputs
 - Large energy gap (MSD=3.8127): significant output magnitude differences
 - Minor distribution shift (KS=0.149): subtle statistical differences
-

epoch_3_013026 vs epoch_5_013026

Head	Correlation	MSD	Var Ratio	KS Stat	Wasserstein
call_offset	-0.036	0.1819	0.683	0.110	0.0498
put_offset	-0.038	0.1759	0.637	0.148	0.0502
wing_width	-0.038	0.8435	0.802	0.117	0.0957
dte	-0.037	38.3062	0.750	0.121	0.6352
pop	-0.034	0.0004	6.257	0.221	0.0079
roi	-0.040	0.0121	0.816	0.158	0.0100
max_loss	-0.041	0.0340	1.267	0.113	0.0166
confidence	-0.035	0.0049	0.753	0.128	0.0148
entry	0.012	0.4367	0.519	0.237	0.2228
exit	-0.044	0.4820	1.314	0.116	0.1113

Physics Interpretation:

- Low coherence (avg correlation=-0.033): models responding differently to same inputs
- Large energy gap (MSD=4.0478): significant output magnitude differences
- Minor distribution shift (KS=0.147): subtle statistical differences

epoch_4_013026 vs epoch_5_013026

Head	Correlation	MSD	Var Ratio	KS Stat	Wasserstein
call_offset	-0.023	0.2142	1.009	0.034	0.0090
put_offset	-0.021	0.2102	0.991	0.203	0.0147
wing_width	-0.022	0.9089	0.973	0.031	0.0203
dte	-0.023	42.7409	0.979	0.130	0.1382
pop	0.009	0.0001	0.886	0.117	0.0016
roi	-0.018	0.0126	0.941	0.047	0.0027
max_loss	-0.020	0.0290	0.984	0.037	0.0027
confidence	0.011	0.0051	0.946	0.129	0.0021
entry	-0.002	0.4961	0.921	0.031	0.0240
exit	0.011	0.3829	0.986	0.037	0.0244

Physics Interpretation:

- Low coherence (avg correlation=-0.010): models responding differently to same inputs
- Large energy gap (MSD=4.5000): significant output magnitude differences
- Stable amplitude (variance ratio=0.96): similar output volatility

- Minor distribution shift (KS=0.080): subtle statistical differences

6. Gradient Saliency Analysis

epoch_3_013026

All Features (Gradient Saliency):

Rank	Feature	Saliency
1	ivr	0.2885
2	gap_risk_score	0.2620
3	te	0.2299
4	atr_pct	0.2008
5	theta	0.1735
6	delta	0.1734
7	vol_ewma	0.1687
8	psar_adaptive	0.1656
9	high	0.1603
10	breakout_score	0.1572
11	exec_allow	0.1326
12	pressure_up	0.1305
13	low	0.1292
14	psar_trend	0.1251
15	open	0.1159
16	mtf_consensus	0.1057
17	strike	0.1034
18	bb_mu_dyn	0.1031
19	consolidation_score	0.1022
20	stoch_k_dyn	0.1020
21	cmf	0.1019
22	bw_expansion_rate	0.1006
23	plus_di	0.0972
24	rsi_dyn	0.0968
25	pressure_down	0.0911
26	adx_adaptive	0.0906
27	bb_lower_dyn	0.0905
28	ret_z	0.0895

Rank	Feature	Saliency
29	bb_sigma_dyn	0.0889
30	close	0.0873
31	position_size_mult	0.0871
32	bb_upper_dyn	0.0814
33	kappa_proxy	0.0799
34	fuzzy_reversion_11	0.0793
35	iv_confidence	0.0779
36	spread_ratio	0.0761
37	log_return	0.0754
38	macd_norm	0.0750
39	macd_histogram	0.0748
40	friction_ratio	0.0736
41	gamma	0.0730
42	psar_reversion_mu	0.0714
43	iv	0.0698
44	target_spot	0.0674
45	bb_percentile	0.0671
46	chaos_membership	0.0667
47	max_dd_60m	0.0660
48	minus_di	0.0660
49	beta1_norm_stub	0.0647
50	risk_override	0.0644
51	vega	0.0625
52	volume	0.0600
53	macd_signal_norm	0.0531
54	vol_energy	0.0504

Temporal Focus: Recent 20% weight=0.0095, Distant 20% weight=0.4441

epoch_4_013026

All Features (Gradient Saliency):

Rank	Feature	Saliency
1	gap_risk_score	0.1502
2	te	0.1403

Rank	Feature	Saliency
3	ivr	0.1313
4	vol_ewma	0.1122
5	theta	0.1030
6	delta	0.1010
7	breakout_score	0.0948
8	atr_pct	0.0937
9	strike	0.0802
10	pressure_up	0.0753
11	exec_allow	0.0728
12	high	0.0712
13	low	0.0660
14	consolidation_score	0.0646
15	pressure_down	0.0634
16	bb_upper_dyn	0.0631
17	bb_mu_dyn	0.0629
18	psar_adaptive	0.0605
19	chaos_membership	0.0553
20	stoch_k_dyn	0.0553
21	position_size_mult	0.0547
22	rsi_dyn	0.0533
23	gamma	0.0518
24	psar_trend	0.0512
25	mtf_consensus	0.0510
26	cmf	0.0508
27	vega	0.0506
28	open	0.0484
29	iv_confidence	0.0475
30	bb_sigma_dyn	0.0474
31	risk_override	0.0468
32	iv	0.0453
33	bw_expansion_rate	0.0449
34	adx_adaptive	0.0448
35	target_spot	0.0444
36	psar_reversion_mu	0.0441
37	max_dd_60m	0.0439
38	spread_ratio	0.0438
39	beta1_norm_stub	0.0438

Rank	Feature	Saliency
40	fuzzy_reversion_11	0.0434
41	bb_lower_dyn	0.0433
42	close	0.0425
43	ret_z	0.0424
44	volume	0.0421
45	plus_di	0.0403
46	macd_histogram	0.0393
47	minus_di	0.0385
48	macd_norm	0.0381
49	friction_ratio	0.0371
50	bb_percentile	0.0366
51	kappa_proxy	0.0355
52	vol_energy	0.0340
53	macd_signal_norm	0.0336
54	log_return	0.0306

Temporal Focus: Recent 20% weight=0.0071, Distant 20% weight=0.2203

epoch_5_013026

All Features (Gradient Saliency):

Rank	Feature	Saliency
1	gap_risk_score	0.3113
2	ivr	0.3084
3	te	0.2822
4	vol_ewma	0.2467
5	atr_pct	0.2350
6	theta	0.2272
7	delta	0.2077
8	high	0.1985
9	exec_allow	0.1871
10	breakout_score	0.1830
11	pressure_up	0.1634
12	low	0.1537
13	strike	0.1447

Rank	Feature	Saliency
14	consolidation_score	0.1427
15	psar_adaptive	0.1415
16	mtf_consensus	0.1233
17	stoch_k_dyn	0.1198
18	bb_lower_dyn	0.1195
19	open	0.1194
20	pressure_down	0.1171
21	bb_mu_dyn	0.1165
22	bb_sigma_dyn	0.1162
23	bb_upper_dyn	0.1144
24	close	0.1142
25	rsi_dyn	0.1134
26	position_size_mult	0.1119
27	cmf	0.1056
28	max_dd_60m	0.1026
29	bw_expansion_rate	0.1016
30	adx_adaptive	0.0966
31	spread_ratio	0.0964
32	target_spot	0.0959
33	gamma	0.0957
34	plus_di	0.0940
35	chaos_membership	0.0926
36	vega	0.0923
37	iv	0.0911
38	psar_reversion_mu	0.0899
39	iv_confidence	0.0898
40	fuzzy_reversion_11	0.0877
41	macd_histogram	0.0875
42	beta1_norm_stub	0.0866
43	ret_z	0.0827
44	risk_override	0.0814
45	macd_signal_norm	0.0813
46	psar_trend	0.0809
47	macd_norm	0.0798
48	minus_di	0.0774
49	volume	0.0773
50	friction_ratio	0.0725

Rank	Feature	Saliency
51	bb_percentile	0.0693
52	log_return	0.0689
53	kappa_proxy	0.0668
54	vol_energy	0.0644

Temporal Focus: Recent 20% weight=0.0077, Distant 20% weight=0.5420

Gradient Alignment: epoch_3_013026 vs epoch_4_013026

- High feature attention alignment (0.985): models focus on same features
- Temporal attention aligned (0.997): same recency bias

Gradient Alignment: epoch_3_013026 vs epoch_5_013026

- High feature attention alignment (0.991): models focus on same features
- Temporal attention aligned (0.984): same recency bias

Gradient Alignment: epoch_4_013026 vs epoch_5_013026

- High feature attention alignment (0.994): models focus on same features
 - Temporal attention aligned (0.986): same recency bias
-

7. Hessian & Fisher Analysis

Hessian Eigenspectrum

Fisher Information

epoch_3_013026 vs epoch_4_013026:

- Uneven sensitivity distribution (spread=20.1x): parameter importance concentrated

epoch_3_013026 vs epoch_5_013026:

- Reduced sensitivity in expert_normal (ratio=0.14x): layer may be undertrained
- Uneven sensitivity distribution (spread=34.1x): parameter importance concentrated

epoch_4_013026 vs epoch_5_013026:

- Relatively uniform sensitivity (spread=1.8x)

8. Surrogate Decision Tree Rules

epoch_3_013026

R2 Score: 0.6814

```
|--- ivr <= 0.674
| |--- bb_sigma_dyn <= 0.832
| | |--- ivr <= 0.304
| | | |--- delta <= 0.695
| | | | |--- value: [-0.112]
| | | |--- delta > 0.695
| | | | |--- value: [-0.093]
| | | |--- ivr > 0.304
| | | | |--- friction_ratio <= -0.550
| | | | |--- value: [-0.035]
| | | | |--- friction_ratio > -0.550
| | | | |--- value: [-0.080]
| | |--- bb_sigma_dyn > 0.832
| | |--- ivr <= -0.058
| | | |--- bb_sigma_dyn <= 2.089
| | | | |--- value: [-0.096]
| | | |--- bb_sigma_dyn > 2.089
| | | | |--- value: [-0.049]
| | |--- ivr > -0.058
| | | |--- vol_ewma <= 1.062
| | | | |--- value: [-0.043]
| | | |--- vol_ewma > 1.062
| | | | |--- value: [0.006]
|--- ivr > 0.674
| |--- ivr <= 1.513
| | |--- bb_sigma_dyn <= 1.139
| | | |--- atr_pct <= 0.945
| | | | |--- value: [-0.034]
| | | |--- atr_pct > 0.945
| | | | |--- value: [0.001]
| | |--- bb_sigma_dyn > 1.139
| | | |--- bb_percentile <= 0.502
| | | | |--- value: [0.035]
| | | |--- bb_percentile > 0.502
| | | | |--- value: [0.006]
| |--- ivr > 1.513
| | |--- ivr <= 2.332
| | | |--- bb_sigma_dyn <= -0.373
| | | | |--- value: [-0.002]
| | | |--- bb_sigma_dyn > -0.373
| | | | |--- value: [0.035]
| | |--- ivr > 2.332
| | | |--- bb_sigma_dyn <= 2.203
| | | | |--- value: [0.047]
| | | |--- bb_sigma_dyn > 2.203
| | | | |--- value: [0.067]
```

epoch_4_013026

R2 Score: 0.7477

```
|--- ivr <= 0.687
|   |--- ivr <= 0.152
|   |   |--- delta <= 0.654
|   |   |   |--- log_return <= -1.541
|   |   |   |   |--- value: [-0.087]
|   |   |   |--- log_return > -1.541
|   |   |   |   |--- value: [-0.119]
|   |   |--- delta > 0.654
|   |   |   |--- delta <= 0.879
|   |   |   |   |--- value: [-0.083]
|   |   |   |--- delta > 0.879
|   |   |   |   |--- value: [-0.106]
|   |--- ivr > 0.152
|   |   |--- atr_pct <= 0.338
|   |   |   |--- theta <= 0.009
|   |   |   |   |--- value: [-0.062]
|   |   |   |--- theta > 0.009
|   |   |   |   |--- value: [-0.094]
|   |   |--- atr_pct > 0.338
|   |   |   |--- ivr <= 0.332
|   |   |   |   |--- value: [-0.062]
|   |   |   |--- ivr > 0.332
|   |   |   |   |--- value: [-0.036]
|--- ivr > 0.687
|   |--- ivr <= 1.626
|   |   |--- ivr <= 1.166
|   |   |   |--- atr_pct <= 0.949
|   |   |   |   |--- value: [-0.024]
|   |   |   |--- atr_pct > 0.949
|   |   |   |   |--- value: [0.001]
|   |   |--- ivr > 1.166
|   |   |   |--- atr_pct <= 1.337
|   |   |   |   |--- value: [0.004]
|   |   |   |--- atr_pct > 1.337
|   |   |   |   |--- value: [0.036]
|   |--- ivr > 1.626
|   |   |--- bb_sigma_dyn <= 1.909
|   |   |   |--- vol_ewma <= -0.319
|   |   |   |   |--- value: [0.020]
|   |   |   |--- vol_ewma > -0.319
|   |   |   |   |--- value: [0.054]
|   |   |--- bb_sigma_dyn > 1.909
|   |   |   |--- ivr <= 1.906
|   |   |   |   |--- value: [0.063]
|   |   |   |--- ivr > 1.906
|   |   |   |   |--- value: [0.084]
```

epoch_5_013026

R2 Score: 0.8012

```
|--- ivr <= 0.496
|   |--- ivr <= 0.157
|   |   |--- delta <= 0.650
|   |   |   |--- ivr <= -0.012
|   |   |   |   |--- value: [-0.120]
|   |   |   |--- ivr > -0.012
|   |   |   |   |--- value: [-0.097]
|   |   |--- delta > 0.650
|   |   |   |--- delta <= 0.952
|   |   |   |   |--- value: [-0.092]
|   |   |   |--- delta > 0.952
|   |   |   |   |--- value: [-0.113]
|   |--- ivr > 0.157
|   |   |--- atr_pct <= 0.957
|   |   |   |--- ivr <= 0.277
|   |   |   |   |--- value: [-0.083]
|   |   |   |--- ivr > 0.277
|   |   |   |   |--- value: [-0.053]
|   |   |--- atr_pct > 0.957
|   |   |   |--- value: [-0.031]
|--- ivr > 0.496
|   |--- ivr <= 1.522
|   |   |--- ivr <= 1.270
|   |   |   |--- atr_pct <= 1.346
|   |   |   |   |--- value: [-0.015]
|   |   |   |--- atr_pct > 1.346
|   |   |   |   |--- value: [0.005]
|   |   |--- ivr > 1.270
|   |   |   |--- bb_sigma_dyn <= 1.586
|   |   |   |   |--- value: [0.008]
|   |   |   |--- bb_sigma_dyn > 1.586
|   |   |   |   |--- value: [0.045]
|   |--- ivr > 1.522
|   |   |--- ivr <= 1.732
|   |   |   |--- vol_ewma <= 0.260
|   |   |   |   |--- value: [0.021]
|   |   |   |--- vol_ewma > 0.260
|   |   |   |   |--- value: [0.059]
|   |   |--- ivr > 1.732
|   |   |   |--- atr_pct <= 2.934
|   |   |   |   |--- value: [0.078]
|   |   |   |--- atr_pct > 2.934
|   |   |   |   |--- value: [0.103]
```

9. Technical Assessment: Pros & Cons

epoch_3_013026

Strengths:

- High surrogate fidelity ($R^2=0.681 \geq 0.50$): decision tree rules are reliable summary
- Signal features consistent with peer checkpoints (avg top-10 overlap=0.95)
- Attribution not overly concentrated (top feature=11.9%, entropy=3.30)
- Strong attribution consistency (SHAP vs Permutation top-5 overlap=80%)

Concerns:

- Possible collapse: near-constant outputs detected (variance near zero)
 - Output drift: call_offset correlation=0.01 (below 0.70 vs peer)
 - Output drift: call_offset correlation=-0.04 (below 0.70 vs peer)
-

epoch_4_013026

Strengths:

- High surrogate fidelity ($R^2=0.748 \geq 0.50$): decision tree rules are reliable summary
- Signal features consistent with peer checkpoints (avg top-10 overlap=0.95)
- Attribution not overly concentrated (top feature=16.2%, entropy=3.21)
- Strong attribution consistency (SHAP vs Permutation top-5 overlap=100%)

Concerns:

- Possible collapse: near-constant outputs detected (variance near zero)
 - Output drift: call_offset correlation=0.01 (below 0.70 vs peer)
 - Output drift: call_offset correlation=-0.02 (below 0.70 vs peer)
-

epoch_5_013026

Strengths:

- High surrogate fidelity ($R^2=0.801 \geq 0.50$): decision tree rules are reliable summary
- Signal features consistent with peer checkpoints (avg top-10 overlap=0.90)
- Attribution not overly concentrated (top feature=17.6%, entropy=3.12)
- Strong attribution consistency (SHAP vs Permutation top-5 overlap=80%)

Concerns:

- Possible collapse: near-constant outputs detected (variance near zero)
- Output drift: call_offset correlation=-0.04 (below 0.70 vs peer)
- Output drift: call_offset correlation=-0.02 (below 0.70 vs peer)

10. Final Recommendation & Ranking

Overall Ranking

Rank	Model	Score	Breakdown
1	epoch_3_013026	12	Pros:8, Cons:-3, Interpretability:3, Feature Balance:2, Consistency:2
2	epoch_4_013026	12	Pros:8, Cons:-3, Interpretability:3, Feature Balance:2, Consistency:2
3	epoch_5_013026	12	Pros:8, Cons:-3, Interpretability:3, Feature Balance:2, Consistency:2

Recommendation

Recommended Model: epoch_3_013026

Rationale (All Strengths):

- High surrogate fidelity ($R^2=0.681 \geq 0.50$): decision tree rules are reliable summary
- Signal features consistent with peer checkpoints (avg top-10 overlap=0.95)
- Attribution not overly concentrated (top feature=11.9%, entropy=3.30)
- Strong attribution consistency (SHAP vs Permutation top-5 overlap=80%)

Runner-up: epoch_4_013026

References

1. **Neural CDE Theory:** Kidger et al., *Neural Controlled Differential Equations for Irregular Time Series* (2020)
2. **System Architecture:** docs/scientific_spec.md - Complete mathematical specification
3. **Feature Schema:** V2.2 with 54 input dimensions (OHLCV, Greeks, dynamic indicators)
4. **Loss Function:** Composite Risk-Aligned Loss with Huber, Sharpe proxy, and drawdown penalties

Composite Loss Function (from training):

$$\mathcal{L}_{\text{composite}} = \lambda_1 \mathcal{L}_{\text{pred}} - \lambda_2 \mathcal{L}_{\text{sharpe}} + \lambda_3 \mathcal{L}_{\text{dd}} + \lambda_4 \mathcal{L}_{\text{turn}}$$

Where:

- $\mathcal{L}_{\text{pred}}$: Huber loss for strike predictions
- $\mathcal{L}_{\text{sharpe}}$: Negative Sharpe ratio (maximized via gradient descent)
- $\mathcal{L}_{\text{dd}}$: Soft drawdown penalty
- $\mathcal{L}_{\text{turn}}$: Turnover penalty for position stability

Interpretations:

$\mathcal{L}_{\text{pred}}$ measures the discrepancy between the model's predicted outputs and the true targets. In this system it is implemented as a **Huber loss** on strike and structure-related outputs, balancing robustness to outliers with sensitivity to small errors. This term anchors training to factual market outcomes and provides the primary supervised learning signal.

$\mathcal{L}_{\text{sharpe}}$ penalizes low risk-adjusted returns and rewards stable profitability. It is typically defined as the **negative Sharpe ratio** (or a smooth proxy thereof), so that minimizing the overall loss corresponds to maximizing risk-adjusted performance. This term encourages the model to favor predictions that lead to consistent returns with controlled volatility, rather than high-variance outcomes.

$\mathcal{L}_{\text{dd}}$ penalizes large peak-to-trough losses in the equity curve implied by the model's decisions. It acts as a **risk-control regularizer**, discouraging strategies that achieve returns by tolerating deep or prolonged drawdowns. This term biases training toward capital preservation and smoother equity trajectories, aligning the model with institutional risk constraints.

$\mathcal{L}_{\text{turn}}$ penalizes excessive changes in positions implied by the model's outputs (high trading frequency or large reallocations). It encourages **temporal smoothness and execution stability**, reducing transaction costs, slippage, and market impact. This term enforces practical deployability by aligning the learned strategy with realistic execution and liquidity constraints.

Report generated by audit_cde_comparison.py | Mathematical foundations from docs/scientific_spec.md